Numerical Integration

$$I(f) = \int_a^b f(x)\,dx \longrightarrow P_n(x)$$

Polynomial ফিট integrate করলে approximate value পাই, exact error নয় exact value যায়।

So, $\int_a^b P_n(x)$

$= \int_a^b \sum_{k=0}^{n} l_k(x) + f(x_k)\,dx$

If we use lagrange.

$P_1(x) = l_0(x) f(x_0) + l_1(x) f(x_1)$ ...

$P_n(x) = \sum_{i=0}^{n} l_i(x) f(x_i)$

So, $I_n = \int_a^b \sum_{k=0}^{n} l_k(x) f(x_k)\,dx$

Weight function $\rightarrow \delta_k = \int_a^b l_k(x)\,dx$

If the nodes are equal distant, this formula is called Newton's Cotes formula

$(a, b) \rightarrow$ a, b excluded

$[a, b] \Rightarrow$ a, b included

So,

$$I_n = \int_a^b \sum_{k=0}^{n} l_k(x) f(x_k) \, dx$$

So, $I_n = \sum_{k=0}^{n} \delta_k f(x_k) \, dx$

Closed Newton's Cote's formula

$[a, b]$ Given interval where, $a = x_0$ and $b = x_n$

as, $a = x_0 < x_1 < x_2 < x_3 \ldots < x_{n-1} < x_n = b$

and, $h = \frac{b-a}{n}$

↳ Step size

Open Newton's Cote's formula

$[a, b]$ given interval where $a < x_0$ and $x_n < b$

as, $a < x_0 < x_1 < x_2 < x_3 \ldots < x_{n-1} < x_n < b$

$$h = \frac{b-a}{n+2}$$

Trapizium Rule (closed Newton's Cotes formula)

with $n = 1$ — [So the interpolating polynomial will be of degree 1 and there will be 2 nodes. So, $h = \frac{b-a}{n} = b-a$

if $x \in [a, b]$; $x_0 = a$; $x_1 = b$

We know,

$$P_1(x) = l_0(x) f(x_0) + l_1(x) f(x_1)$$

So,

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x-b}{a-b}$$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x-a}{b-a}$$

Now, $\delta_0 = \int_a^b l_0(x)\, dx$

$$= \int_a^b \frac{x-b}{a-b}\, dx$$

$$= \frac{1}{a-b} \int_a^b (x-b)\, dx = \frac{1}{a-b} \left[\frac{1}{2} x^2 - bx \right]_a^b$$

$$= \frac{b-a}{2}$$

Now, $\delta_1 = \int_a^b l_1(x)\, dx$

$$= \int_a^b \frac{x-a}{b-a}\, dx$$

$$= \frac{1}{b-a}\left[\frac{1}{2}x^2 - ax\right]_a^b$$

$$= \frac{b-a}{2}$$

So, $I_n = \int_a^b P_1(x)\, dx$

$$\Rightarrow I_n = \int_a^b \left(l_0(x) f(x_0) + l_1(x) f_1(x_1)\right) dx$$

$$\Rightarrow I_n = \int_a^b l_0(x) f(x_0)\, dx + \int_a^b l_1(x) f_1(x)\, dx$$

$$\Rightarrow I_n = \delta_0 f(x_0) + \delta_1 f(x_1)$$

$$\therefore \boxed{I_n = \frac{b-a}{2}\left[f(a) + f(b)\right]} \quad \leftarrow n=1$$

Steps → ① Normal Integration using Trapezium Rule
② Actual Value ③ % Error

Ex $[0,2]$ and $f(x) = e^x$

$\int e^{3x} = \frac{1}{3}e^x$

$\int e^{nx} = \frac{1}{n}e^x$

We know, ~~trapizium~~ trapezium formula,

$$I_n = \frac{b-a}{2}\left[f(a) + f(b)\right]$$

$$= \frac{2-0}{2}\left[e^2 + e^0\right]$$

$$= 8.3891 \quad ; \text{ Approximate Value}$$

Now ~~Accurate~~ Actual Value,

$$\int_a^b f(x)\,dx = \int_0^2 e^x\,dx = \left[e^x\right]_0^2$$

$$= 6.3891$$

So, Percentage Error $= \left|\frac{8.3891 - 6.3891}{6.3891}\right| \times 100\%$

$= 31.3\%$

Huge Error

We need to Reduce the Error.

Cauchys Theorem → Upper Bound

Upperbound of interpolation Error using Cauchys Theorem.

$$\left| \frac{f^{(n+1)}\,\xi}{(n+1)!} \right| \overbrace{\left| (x-x_0)(x-x_1)\cdots(x-x_n) \right|}^{\omega(x)}$$

$$\max \xi \in [a,b]$$

Ex → $n=1,\ f(x) = e^x\ ;\ [0,2]$

So, $$\left| \frac{f^{(n+1)}\,\xi}{(n+1)!} \right| \quad \max \in (a,b)$$

$$= \left| \frac{f^2(\xi)}{2!} \right| \quad [\xi \text{ ला max ची value द्यायची}]$$

$$\therefore \frac{1}{2!} e^{(2)}$$

Now, $\omega(x) = (x-x_0)(x-x_1)$ [as $n=1$]

so, $\omega(x) = (x-0)(x-2) = x^2 - 2x$

$$\text{So, } \int_0^2 \omega(x)\,dx = \int_0^2 (x^2 - 2x)\,dx$$

$$= \left[\frac{1}{3}x^3 - \frac{2}{2}x^2\right]_0^2$$

$$= \frac{4}{3}$$

So, Upper bound $= \frac{1}{2!}e^2\left(\frac{4}{3}\right)$

Using Composite → To reduce the percentage error

Composite Newton-Cote formula

This method improves the result without increasing the actual node numbers.

$[a, b]$ into m subintervals. [divide equal widths]

$C_{1,m}$ → Composite Newton's Cote formula.

$$\sum_{i=0}^{m} I_{1,i} = \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \ldots + 2f(x_{n-1}) + f(x_n)\right]$$

h = Width; Gap between 2 nodes.

So, $x_1 = x_0 + h$

[0,2]; $f(x) = e^x$; $m = 2$; Composite.

We know $h = \frac{b-a}{m} = \frac{2-0}{2} = 1$

So, Needed nodes → x_0 (↓ 0), x_1 (↓ x_0+h), x_2 (↓ 2)

So, $\frac{h}{2}\left[f(x_0) + 2f(x_1) + f(x_2)\right]$

$= \frac{1}{2}\left[e^0 + 2 \cdot e^1 + e^2\right]$

$= 6.9128$

So, Percentage Error: $= \left|\frac{6.9128 - 6.3891}{6.3891}\right| \times 100$

$= 8.197\%$

if $m = 4$ $\quad h = \frac{b-a}{m} = \frac{2-0}{4} = 0.5$

So, x_0, x_1, x_2, x_3, x_4; $x_0 = 0$; $x_1 = 0.5$, $x_2 = 1$, $x_3 = 1.5$, $x_4 = 2$.

So, $I_n = \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4)\right]$

$= \frac{0.5}{2}\left[e^0 + e^{0.5} + e^1 + e^{1.5} + e^2\right]$ = 6.522%

Error = 2.08%.

m এর মান বাড়লে, Error Percentage কি পরিমাণ কমে

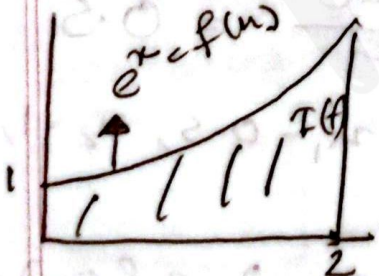
Here, for $m=4$, we have $h=0.5$ and the result is,

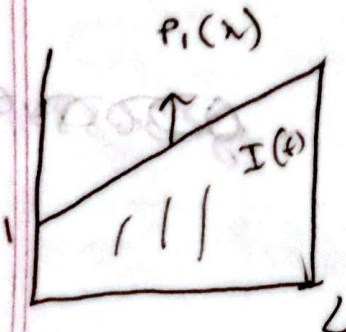
$$C_{1,4} = \frac{0.5}{2}\left(e^0 + 2e^{0.5} + 2e^1 + 2e^{1.5} + e^2\right)$$

$$= 6.522$$

m	h	C_{1m}	$\lvert I-C_{1,m}\rvert$
1	2	8.389	2
2	1	6.913	0.524
3	2/3	6.624	0.235
4	1/2	6.522	0.133
8	0.125	6.422	0.033
16	0.125	6.397	0.008

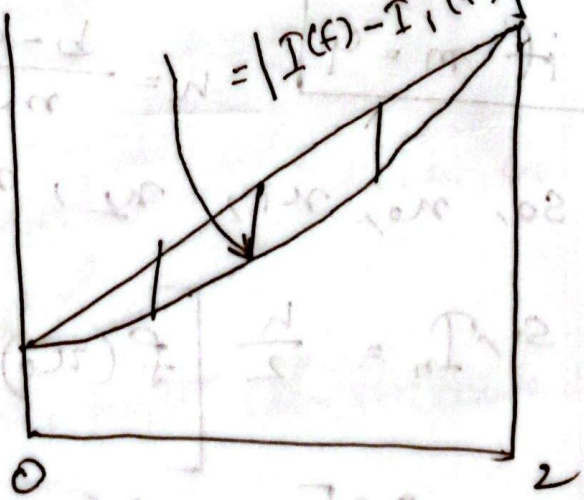
* Here, The errors are decreasing as m increases
* The error decreases by a factor of 4 indicating that it is a quadratic convergence.





Error = area enclosed between the curve

$= |I(f) - I_1(f)|$



Simpson's Rule

Newton's Cotes Formula for $n=1$ (Trapezium Rule)

$n=2$; Simpson's Rule.

for $n=1$,

$$I_1(f) = \int_a^b P_1(x)\,dx \quad \ldots\ldots (i)$$

$$P_1(x) = l_0(x) f(x_0) + l_1(x) f(x_1)$$

$$\int P_1(x) = \int l_0(x) f(x_0)\,dx + \int l_1(x) f(x_1)\,dx$$

$$= f(x_0) \int l_0(x)\,dx + f(x_1) \int l_1(x)\,dx$$

$$= \delta_0 f(x_0) + \delta_1 f(x_1)$$

$$= \frac{b-a}{2} \times \left[f(a) + f(b) \right]$$

So, $$I_1(f) = \frac{b-a}{2} \times \left[f(a) + f(b) \right]$$

$$\sum_{k=0}^{1} \delta_k f(x_k)$$

~~$I_2(f) =$~~

for $n=2$, Simpson's Rule,

$$I_2(f) = \sum_{k=0}^{2} a_k f(x_k)$$

Now, the lagrance basis will be)

$$l_0(x) = \left(\frac{x-x_1}{x_0-x_1}\right)\left(\frac{x-x_2}{x_0-x_2}\right)$$

$$= \frac{(x-m)(x-b)}{(a-m)(a-b)}$$

We know, $h = \frac{b-a}{2}$

For intervals, $[a,b]$ for $n=2$ we get 3 intervals, x_0, x_1, x_2.

Here, $x_0 = a$, $x_2 = b$

So, $x_1 = a+h$

$$= a + \frac{b-a}{2}$$

$$= \frac{a+b}{2} = m$$

$$l_1(x) = \left(\frac{x-x_0}{x_1-x_0}\right)\left(\frac{x-x_2}{x_1-x_2}\right)$$

$$= \frac{(x-a)(x-b)}{(m-a)(m-b)}$$

$$l_2(x) = \left(\frac{x-x_0}{x_2-x_0}\right)\left(\frac{x-x_1}{x_2-x_1}\right)$$

$$= \frac{(x-a)(x-m)}{(b-a)(b-m)}$$

Now,

$$\delta_0 = \int_a^b l_0(x)\,dx$$

$$= \frac{1}{(a-m)(a-b)} \int_a^b (x-m)(x-b)\,dx$$

$$= \frac{1}{6}(b-a)$$

$$\delta_1 = \int_a^b l_1(x)\,dx$$

$$= \frac{1}{(m-a)(m-b)} \int_a^b (x-a)(x-b)\,dx$$

$$= \frac{2}{3}(b-a)$$

$$\delta_2 = \int_a^b l_2(x)\,dx$$

$$= \frac{1}{(b-a)(b-m)} \int_a^b (x-a)(x-m)\,dx$$

$$= \frac{1}{6}(b-a)$$

$$\therefore I_2(f) = \delta_0 f(x_0) + \delta_1 f(x_1) + \delta_2 f(x_2)$$

$$\therefore I_2(f) = \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

Exactness

$\xi \in [a,b]$

We know,

(i) Upper Bound $= \left| \frac{1}{(n+1)!} f^{(n+1)}(\xi)(x-x_0)(x-x_1)\cdots(x-x_n) \right|$;

(ii) Clearly, the error is zero(0) if $f^{(n+1)} = 0$

(iii) In that case, the Newton-Cotes formula will give exact answers.

(iv) so, approximate value = Actual value.

(v) So no more error.

(vi) This also means that the function $f(x)$ is an n-degree polynomial, $P_n(x)$.;

(vii) $f(x) = P_n(x)$

(viii) The above statement implies that the trapezium rule $I_1(f)$ is exact for all functions $f(x) = P_1(x)$

(ix) Definition → The degree of exactness of a quadrature formula is the largest integer n for which the formula is exact for all polynomial $P_n(x)$

(x) Example → (Simpson's Rule) for $n=2$, the Newton-Cotes formula becomes,

$$I_2(f) = \sum_{k=0}^{2} \delta_k f(x_k)$$